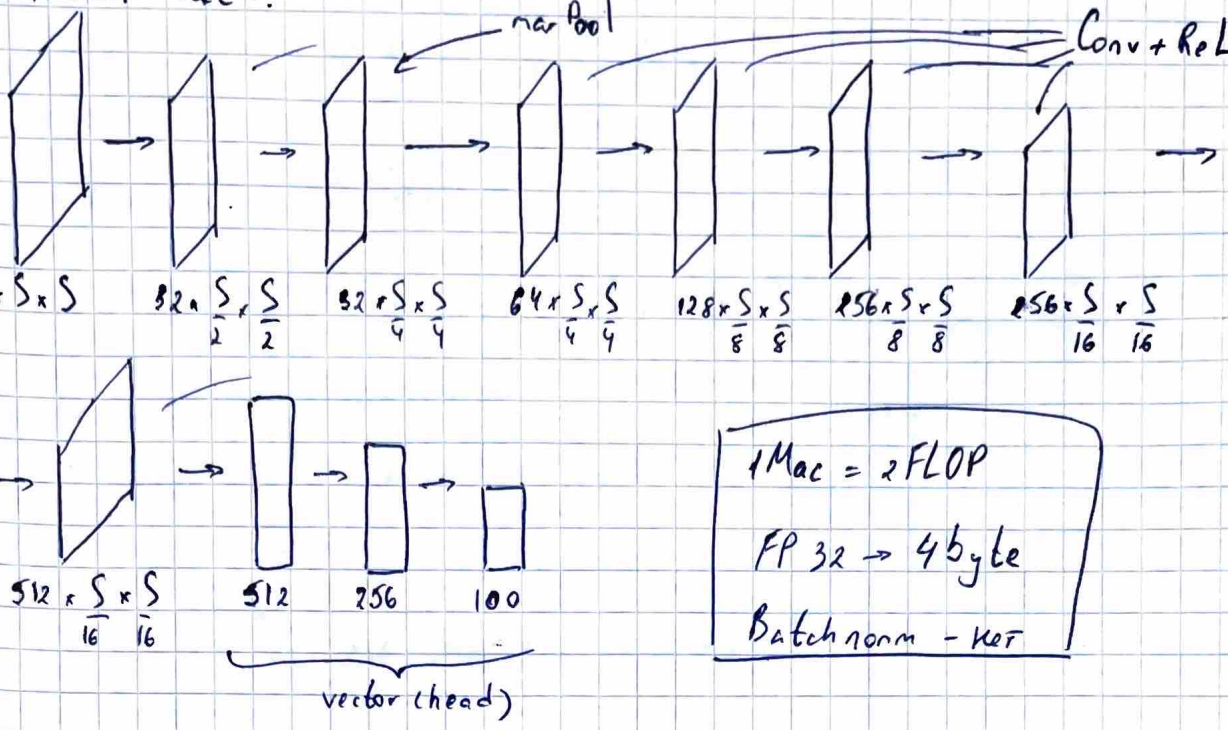
Efficient NN

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Model arch.



1 Mac = 2 FLOP

FP 32 → 4 byte

Batch norm - нет

1) Features map sizes

for conv with padding $= \frac{k}{2}$, odd k, even H: $H_{out} = \left\lfloor \frac{H + 2\cdot\lfloor \frac{k}{2} \rfloor - k}{s} \right\rfloor + 1 = \frac{H}{s}$

Maxpool 3×3, s2, p1: $\left\lfloor \frac{\frac{S}{2} + 2 - 3}{2} \right\rfloor + 1 = \frac{S}{4}$ (т.к. $\frac{S}{2}$ - even)

$3 \times S \times S \xrightarrow{C_1} 32 \times \frac{S}{2} \times \frac{S}{2} \xrightarrow{Pool} 32 \times \frac{S}{4} \times \frac{S}{4} \xrightarrow{C_2} 64 \times \frac{S}{4} \times \frac{S}{4} \xrightarrow{C_3} 128 \times \frac{S}{8} \times \frac{S}{8}$

$\xrightarrow{C_4} 256 \times \frac{S}{8} \times \frac{S}{8} \xrightarrow{C_5} 256 \times \frac{S}{16} \times \frac{S}{16} \xrightarrow{C_6} 512 \times \frac{S}{16} \times \frac{S}{16} \xrightarrow{GAP} 512 \rightarrow$

$256 \rightarrow 100$

number of elem / image: $x: 3S^2$, $a_1: 8S^2$, $p: 2S^2$, $a_2: 4S^2$, $a_3: 2S^2$, $a_4: 4S^2$, $a_5: S^2$, $a_6: 2S^2$,

where
- x - input image
- a_1 - Conv1 + Relu output
- p - maxpool output
- a_2 - conv 2 output
- ...
- a_6 - conv 6 output

2) Flops

$Flops_{conv} = 2 \cdot B \cdot H_{out} \cdot W_{out} \cdot C_{out} \cdot C_{in} \cdot k^2$

$Flops_{lin} = 2 \cdot B \cdot d_{in} \cdot d_{out}$

$F_1 = 2B \cdot \frac{S^2}{4} \cdot 32 \cdot 3 \cdot 49 = 2352\ BS^2$

$F_2 = 2B \cdot \frac{S^2}{16} \cdot 64 \cdot 32 \cdot 25 = 6400\ BS^2$

$F_3 = 2B \cdot \frac{S^2}{64} \cdot 128 \cdot 64 \cdot 9 = 2304\ BS^2$

$F_4 = 2B \cdot \frac{S^2}{64} \cdot 256 \cdot 128 = 1024\ BS^2$

$F_5 = 2B \cdot \frac{S^2}{256} \cdot 256 \cdot 256 \cdot 9 = 4608\ BS^2$

$F_6 = 2B \cdot \frac{S^2}{256} \cdot 512 \cdot 256 = 1024\ BS^2$

$F_{fc1} = 2B \cdot 512 \cdot 256 = 262\,144\ B$

$F_{fc2} = 2B \cdot 256 \cdot 100 = 51\,200\ B$

ReLu, MaxPool + Gap add $\approx 39\ BS^2$ operations $\approx 0{,}2\%$, ignore

$FLOPs(S, B) = B(17712\ S^2 + 313\,344)$

3) Number of params

$N_p = \underbrace{4704}_{C_1} + \underbrace{51\,200}_{C_2} + \underbrace{73\,728}_{C_3} + \underbrace{32\,768}_{C_4} + \underbrace{589\,824}_{C_5} + \underbrace{131\,072}_{C_6} + \underbrace{131\,328}_{fc_1} + \underbrace{25\,700}_{fc_2} = 1\,040\,324$

$W = 4N_p = 4\,161\,296$ bytes

4) Memory

Conv 1 : $4(3S^2 + 8S^2) = 44S^2$, index of MaxPool

MaxPool : $4(3S^2 + 8S^2 + 2S^2) + (8 \cdot 2S^2) = 68S^2$ – max

Conv 2 : $4(3S^2 + 2S^2 + 4S^2) = 36S^2$

Conv 3, Conv 4 : $36S^2$ Conv 5 : $32S^2$ Conv 6 : $24S^2$

Memory $(S, B) = W + 68 B S^2$ (bytes),

where $W = 4\,161\,296$

Out of Memory if $\text{Memory}(S,B) > \text{Memory}(\text{GPU})$

$\text{Memory}(512, 256) \approx 4.57$ GB $<$ 12 GB (on my GPU RTX 4070)

5) Bytes moved (DRAM Traffic)

C_1: $4(3+8) = 44$ $\quad$ ReLU_1: $4(8+8) = 64$

MaxPool: $4(8+2) + 8 \cdot 2 = 56$

C_2: $4(2+4) = 24$ $\quad$ ReLU_2: 32

C_3: $4(4+2) = 24$ $\quad$ ReLU_3: 16

C_4: $4(2+4) = 24$ $\quad$ ReLU_4: 32

C_5: $4(4+1) = 20$ $\quad$ ReLU_5: 8

C_6: $4(1+2) = 12$ $\quad$ ReLU_6: 16

Gap: $4 \cdot 2 = 8$

$\Sigma = 380\ S^2$ bytes/image

Head (Gap, output, fc1, ReLU, fc2): 8592 bytes/image

~~Bytes~~ Weights read once per forward pass: W

$\text{Bytes}(S, B) = 380 B S^2 + 8592 B + W$

Arithmetic Intensity: $AI = \frac{\text{FLOPs}}{\text{Bytes}} \xrightarrow{BS^2 \to \infty} \frac{17\,712}{380} \approx 47$ FLOP/byte

For RTX 4070 12 GB: $AI_{ridge} = \frac{P_{peak}}{BW_{peak}} = \frac{29 \text{ TFLOP/s}}{504 \text{ GB/s}} = 58$ Flops/byte

AI per layer: $AI_i = \frac{F_i}{M_i}$ (activations only)

C_1: $\frac{2352}{44} \approx 53$, $\quad C_2$: $\frac{6400}{24} \approx 267$, $\quad C_3$: $\frac{2304}{24} = 96$,

C_4: $\frac{1024}{24} \approx 43$, $\quad C_5$: $\frac{4608}{20} \approx 230$, $\quad C_6$: $\frac{1024}{12} \approx 85$

C_2, C_3, C_5, C_6 : $AI_i > 58$ → compute bound

C_1, C_4 : $AI_i < 58$ - memory bound

ReLu, MaxPool : $AI \approx 0,1$ - memory bound

Whole Network : $AI \approx 47 < 58$

Small S, B : $AI \approx 4$ dominated by reading the weights - mem. bound.
32^2 1

6) Latency

forward pass - sequence of $N_k = 17$ kernels (6 Conv + 6 ReLu + MaxPool + GAP + fc1 + ReLu + fc2). The time of kernel „i" is bounded from below by the launch cost „$\hat{\tau}$".

$$\text{Latency}(S, B, \Theta) = \sum_{i=1}^{N_k} \max\left(\hat{\tau}, \frac{F_i}{P}, \frac{M_i}{BW}\right), \quad \Theta = (\tau, P, BW)$$

where F_i, M - FLOPs and bytes of kernel i

τ - per-kernel launch overhead [s]

P - attainable compute throughput [FLOPs/s]

BW - attainable memory band with [bytes/s]

Assumption: FP32 without TF32; cudnn.benchmark = False, one value of P for all convs kernels execute sequent., L2 cache (36 MB) - ignored, although for small S, B weights activations fit into it; GPU clock is const.

7) Energy

$$\text{Energy}(S, B, \Theta_E) = P_s \cdot \text{Latency}(S, B, \Theta) + e_F \cdot \text{FLOPs}(S, B) + e_M \cdot \text{Bytes}(S, B)$$

$\Theta_E = (P_s, e_F, e_M)$: P_s - static power [W], e_F - energy per FLOP [J]

e_M - energy per DRAM [J] Assumption: the whole board is measurable energy of 1 forw. pass = counter diff over N runs / by N; P_s doesn't depend on load, no power throttling (RTX 4070 TDP 200 W)

Answer

$$\text{FLOPs}(S, B) = B(17\,712\,S^2 + 313\,344)$$

$$\text{Memory}(S, B) = 68\,BS^2 + 4\,161\,296 \text{ [bytes]}$$

$$\text{Latency}(S, B, \Theta) = \sum_{i=1}^{17} \max\left(\hat{\tau}, \frac{F_i}{P}, \frac{M_i}{BW}\right)$$

$$\text{Energy}(S, B, \Theta_E) = P_s \cdot \text{Latency} + e_F \cdot \text{FLOPs} + e_M \cdot \text{Bytes}$$